

未命名 1

1 Kinematics

$$\begin{aligned}\bar{u}_\alpha &= u_\alpha + x_3 \phi_\alpha, & \bar{u}_3 &= w, \\ \epsilon_{\alpha\beta} &= \varepsilon_{\alpha\beta} + x_3 \kappa_{\alpha\beta}, \\ \varepsilon_{\alpha\beta} &= \frac{1}{2}(u_{\alpha,\beta} + u_{\beta,\alpha}), & \kappa_{\alpha\beta} &= -\frac{1}{2}(\phi_{\alpha,\beta} + \phi_{\beta,\alpha}), \\ \gamma_\alpha &= w_{,\alpha} - \phi_\alpha.\end{aligned}$$

2 Constitutive relations

$$\begin{aligned}\sigma_{\alpha\beta} &= D_{\alpha\beta\gamma\eta} \epsilon_{\gamma\eta}, & \tau_{3\alpha} &= G \gamma_\alpha, \\ D_{\alpha\beta\gamma\eta} &= \frac{E}{1-\nu^2} \left(\nu \delta_{\alpha\beta} \delta_{\gamma\eta} + \frac{1-\nu}{2} (\delta_{\alpha\gamma} \delta_{\beta\eta} + \delta_{\alpha\eta} \delta_{\beta\gamma}) \right), \\ G &= \frac{E}{2(1+\nu)}.\end{aligned}$$

3 Stress resultants

$$\begin{aligned}\bar{N}_{\alpha\beta} &= \int_{-h/2}^{h/2} \sigma_{\alpha\beta} dx_3 = h D_{\alpha\beta\gamma\eta} \varepsilon_{\gamma\eta}, \\ M_{\alpha\beta} &= \int_{-h/2}^{h/2} x_3 \sigma_{\alpha\beta} dx_3 = \frac{h^3}{12} D_{\alpha\beta\gamma\eta} \kappa_{\gamma\eta}, \\ Q_\alpha &= k \int_{-h/2}^{h/2} \sigma_{3\alpha} dx_3 = k G h \gamma_\alpha.\end{aligned}$$

4 Equilibrium equations and boundary conditions (strong form)

$$\delta\Pi = \int_{\Omega} \left(\bar{N}_{\alpha\beta} \delta\varepsilon_{\alpha\beta} + M_{\alpha\beta} \delta\kappa_{\alpha\beta} + Q_\alpha \gamma_\alpha - N_{\alpha\beta} w_{,\alpha} \delta w_{,\beta} \right) d\Omega = 0.$$

Integration by parts yields:

$$\begin{aligned}\bar{N}_{\alpha\beta,\beta} &= 0 \quad \text{in } \Omega, \\ M_{\alpha\beta,\beta} - Q_\alpha &= 0 \quad \text{in } \Omega, \\ Q_{\alpha,\alpha} - (N_{\alpha\beta} w_{,\beta})_{,\alpha} &= 0 \quad \text{in } \Omega.\end{aligned}$$

Natural boundary conditions:

$$\bar{N}_{\alpha\beta} n_\beta = \bar{N}_\alpha \quad \text{on } \Gamma_N, \quad M_{\alpha\beta} n_\beta = \bar{M}_\alpha \quad \text{on } \Gamma_M, \quad (Q_\alpha - N_{\alpha\beta} w_{,\beta}) n_\alpha = \bar{Q} \quad \text{on } \Gamma_Q.$$

5 Galerkin weak form

Multiply each strong form equation by the corresponding test function δu_α , $\delta \phi_\alpha$, δw and integrate over Ω :

$$\int_{\Omega} N_{\alpha\beta,\beta} \delta u_\alpha d\Omega = 0, \quad \int_{\Omega} (M_{\alpha\beta,\beta} - Q_\alpha) \delta \phi_\alpha d\Omega = 0,$$

$$\int_{\Omega} (Q_{\alpha,\alpha} - (P_{\alpha\beta} w_{,\beta})_{,\alpha}) \delta w d\Omega = 0.$$

Integrate by parts to transfer derivatives to the test functions:

$$\int_{\Omega} N_{\alpha\beta,\beta} \delta u_\alpha d\Omega = \int_{\Gamma} N_{\alpha\beta} n_\beta \delta u_\alpha d\Gamma - \int_{\Omega} N_{\alpha\beta} \delta u_{\alpha,\beta} d\Omega = 0.$$

$$\int_{\Omega} M_{\alpha\beta,\beta} \delta \phi_\alpha d\Omega - \int_{\Omega} Q_\alpha \delta \phi_\alpha d\Omega = \int_{\Gamma} M_{\alpha\beta} n_\beta \delta \phi_\alpha d\Gamma - \int_{\Omega} M_{\alpha\beta} \delta \phi_{\alpha,\beta} d\Omega - \int_{\Omega} Q_\alpha \delta \phi_\alpha d\Omega = 0.$$

$$\int_{\Omega} Q_{\alpha,\alpha} \delta w d\Omega - \int_{\Omega} (P_{\alpha\beta} w_{,\beta})_{,\alpha} \delta w d\Omega = \int_{\Gamma} Q_\alpha n_\alpha \delta w d\Gamma - \int_{\Omega} Q_\alpha \delta w_{,\alpha} d\Omega$$

$$- \int_{\Gamma} P_{\alpha\beta} w_{,\beta} n_\alpha \delta w d\Gamma + \int_{\Omega} P_{\alpha\beta} w_{,\beta} \delta w_{,\alpha} d\Omega = 0.$$

Move all domain integrals to one side and boundary integrals to the other to obtain the weak form:

$$\int_{\Omega} (M_{\alpha\beta} \delta \phi_{\alpha,\beta} + Q_\alpha \delta \phi_\alpha + Q_\alpha \delta w_{,\alpha} - P_{\alpha\beta} w_{,\beta} \delta w_{,\alpha}) d\Omega = \int_{\Gamma_M} \bar{M}_\alpha \delta \phi_\alpha d\Gamma + \int_{\Gamma_Q} \bar{Q} \delta w d\Gamma.$$

Using the symmetry of strain and curvature:

$$N_{\alpha\beta} \delta u_{\alpha,\beta} = N_{\alpha\beta} \delta \varepsilon_{\alpha\beta}, \quad M_{\alpha\beta} \delta \phi_{\alpha,\beta} = M_{\alpha\beta} \delta \kappa_{\alpha\beta},$$

Also, $Q_\alpha \delta \phi_\alpha + Q_\alpha \delta w_{,\alpha} = Q_\alpha \delta \gamma_\alpha$, where $\gamma_\alpha = w_{,\alpha} - \phi_\alpha$. If external boundary forces are neglected (pure buckling equilibrium), the weak form coincides with the variational formulation:

$$\boxed{\int_{\Omega} (M_{\alpha\beta} \delta \kappa_{\alpha\beta} + Q_\alpha \delta \gamma_\alpha - P_{\alpha\beta} w_{,\alpha} \delta w_{,\beta}) d\Omega = 0.}$$

The bilinear forms are defined as follows:

$$a_b = \int_{\Omega} \frac{h^3}{12} D_{\alpha\beta\gamma\eta} \kappa_{\alpha\beta} \delta \kappa_{\gamma\eta} d\Omega \quad a_s = \int_{\Omega} k G h \gamma_\alpha \delta \gamma_\alpha d\Omega \quad a_g = \int_{\Omega} P_{\alpha\beta} w_{,\alpha} \delta w_{,\beta} d\Omega$$

$$a_b + a_s - \lambda a_g = 0$$

6 Finite element discretization

$$u_\alpha^h = \sum_I N_I u_{\alpha I} \quad w^h = \sum_I N_I w_I \quad \phi_\alpha^h = \sum_I N_I \phi_{\alpha I}$$

$$\varepsilon_{\alpha\beta I} = \frac{1}{2}(N_{I,\beta}u_{\alpha I} + N_{I,\alpha}u_{\beta I}) \quad \kappa_{\alpha\beta I} = -\frac{1}{2}(N_{I,\beta}\phi_{\alpha I} + N_{I,\alpha}\phi_{\beta I})$$

$$\gamma_{\alpha I} = N_{I,\alpha}w_I - N_I\phi_{\alpha I} \quad w_{,\alpha I} = N_{I,\alpha}w_I$$

$$a_b = \int_{\Omega} \frac{h^3}{12} D_{\alpha\beta\gamma\eta} \kappa_{\alpha\beta}^h \delta \kappa_{\gamma\eta}^h d\Omega$$

$$a_b = \sum_{I,J} \delta \phi_{iI} \left(\int_{\Omega} \frac{h^3}{48} D_{\alpha\beta\gamma\eta} (N_{I,\beta} \delta_{\alpha i} + N_{I,\alpha} \delta_{\beta i}) (N_{J,\eta} \delta_{\gamma j} + N_{J,\gamma} \delta_{\eta j}) d\Omega \right) \phi_{jJ}$$

$$K_{IJ}^b = \frac{h^3}{48} \int_{\Omega} D_{\alpha\beta\gamma\eta} (N_{I,\beta} \delta_{\alpha i} + N_{I,\alpha} \delta_{\beta i}) (N_{J,\eta} \delta_{\gamma j} + N_{J,\gamma} \delta_{\eta j}) d\Omega$$

$$a_s = \int_{\Omega} kGh \gamma_{\alpha}^h \delta \gamma_{\alpha}^h d\Omega$$

$$a_s = \sum_{I,J} \left[\delta w_I \left(\int_{\Omega} kGh N_{I,\alpha} N_{J,\alpha} d\Omega \right) w_J + \delta w_I \left(\int_{\Omega} -kGh N_{I,\alpha} N_J d\Omega \right) \phi_{\alpha J} \right. \\ \left. + \delta \phi_{\alpha I} \left(\int_{\Omega} -kGh N_I N_{J,\alpha} d\Omega \right) w_J + \delta \phi_{\alpha I} \left(\int_{\Omega} kGh N_I N_J \delta_{\alpha\beta} d\Omega \right) \phi_{\beta J} \right]$$

$$K_{IJ}^s = kGh \int_{\Omega} \begin{bmatrix} N_{I,\alpha} N_{J,\alpha} & -N_{I,\alpha} N_J \\ -N_I N_{J,\beta} & N_I N_J \delta_{\alpha\beta} \end{bmatrix} d\Omega$$

$$a_s = \sum_{I,J} [\delta w_I \quad \delta \phi_{\alpha I}] K_{IJ}^s \begin{bmatrix} w_J \\ \phi_{\beta J} \end{bmatrix}$$

$$a_g = \int_{\Omega} P_{\alpha\beta} w_{,\alpha} \delta w_{,\beta} d\Omega$$

$$a_g = \sum_{I,J}^{n_p} \delta w_I \left(\int_{\Omega} P_{\alpha\beta} N_{I,\alpha} N_{J,\beta} d\Omega \right) w_J$$

$$K_{IJ}^g = \int_{\Omega} P_{\alpha\beta} N_{I,\alpha} N_{J,\beta} d\Omega.$$

$$\left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & \mathbf{K}_{ww}^s & \mathbf{K}_{w\phi}^s \\ 0 & \mathbf{K}_{\phi w}^s & \mathbf{K}^b + \mathbf{K}_{\phi\phi}^s \end{bmatrix} - \lambda \begin{bmatrix} 0 & 0 & 0 \\ 0 & \mathbf{K}_{ww}^g & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) \begin{bmatrix} \mathbf{d}_u \\ \mathbf{d}_w \\ \mathbf{d}_\phi \end{bmatrix} = \mathbf{0}$$